CHAPTER 28

Hepatobiliary diseases

Liver cirrhosis

- Cirrhosis is defined histologically as a diffuse hepatic process characterized by fibrosis and conversion of the normal liver architecture into structurally abnormal nodules.
- The progression of liver injury to cirrhosis may occur over several weeks to years.

Signs and symptoms

- Some patients with cirrhosis are completely asymptomatic and have a reasonably normal life expectancy.
- Other individuals have a multitude of the most severe symptoms of end-stage liver disease and a limited chance for survival.
- Common signs and symptoms may stem from decreased hepatic synthetic function (eg, coagulopathy), portal hypertension (eg, variceal bleeding), or decreased detoxification capabilities of the liver (eg, hepatic encephalopathy).

Portal hypertension

- Portal hypertension can have prehepatic, intrahepatic, or posthepatic causes.

Budd-Chiari syndrome, a posthepatic cause, is characterized by the following symptoms:

- Hepatomegaly
- Abdominal pain

Ascites

Ascites is suggested by the following findings on physical examination:

- Abdominal distention
- Bulging flanks
- Shifting dullness
- Elicitation of a "puddle sign" in patients in the knee-elbow position

Hepatic encephalopathy

The symptoms of hepatic encephalopathy may range from mild to severe and may be observed in as many as 70% of patients with cirrhosis. Symptoms are graded on the following scale:

Grade 0 - Subclinical; normal mental status but minimal changes in memory, concentration, intellectual function, coordination

Grade 1 - Mild confusion, euphoria or depression, decreased attention, slowing of ability to perform mental tasks, irritability, disorder of sleep pattern (ie, inverted sleep cycle)

Grade 2 - Drowsiness, lethargy, gross deficits in ability to perform mental tasks, obvious personality changes, inappropriate behavior, intermittent disorientation (usually with regard to time)

Grade 3 - Somnolent, but arousable state; inability to perform mental tasks; disorientation with regard to time and place; marked confusion; amnesia; occasional fits of rage; speech is present but incomprehensible

Grade 4 - Coma, with or without response to painful stimuli

- Findings on physical examination in hepatic encephalopathy include asterixis and fetor hepaticus.

Additional signs and symptoms

- Many patients with cirrhosis experience fatigue, anorexia, weight loss, and muscle wasting.
- Cutaneous manifestations of cirrhosis include jaundice, spider angiomata, skin telangiectasias ("paper money skin"), palmar erythema, white nails, disappearance of lunulae, and finger clubbing, especially in the setting of hepatopulmonary syndrome.

Ancillary investigations

Hepatorenal syndrome

- Hepatorenal syndrome is diagnosed when a creatinine clearance rate of less than 40 mL/

min is present or when a serum creatinine level of greater than 1.5 mg/dL, a urine volume of less than 500 mL/day, and a urine sodium level of less than 10 mEq/L are present.

- Urine osmolality is greater than plasma osmolality.

Portal hypertension

- During angiography, a catheter is placed selectively via either the transjugular or transfemoral route into the hepatic vein to measure portal pressure.

Hepatic encephalopathy

- An elevated arterial or free venous serum ammonia level is the classic laboratory abnormality reported in patients with hepatic encephalopathy.
- Electroencephalography may be helpful in the initial workup of a patient with cirrhosis and altered mental status, when ruling out seizure activity may be necessary.
- Computed tomography (CT) scanning and MRI studies of the brain may be important in ruling out intracranial lesions when the diagnosis of hepatic encephalopathy is in question.

Ascites

- Paracentesis is essential in determining whether ascites is caused by portal hypertension or by another process.

Alcoholic Liver Disease

Alcohol is one of the most common causes of chronic liver disease. In the UK mortality from alcoholic liver disease is rising, with over 3000 deaths / year; the mean age at presentation is falling.

The risk alcoholic liver disease is variable and not everyone who drinks heavily will develop liver disease. Only 10% of alcoholics have evidence of cirrhosis at postmortem. Alcoholic liver disease does not occur below a threshold of 21 units/week in women and 28 units/week in men. Most individuals with liver disease will have drunk heavily for more than 5 years.

Although the average alcohol consumption of an individual with cirrhosis is 160 g/day for an average of 8 years, there is no clear linear relationship between done and liver damage.

Some of the risk factors for alcoholic liver disease are :

- Drinking patterns - The type of beverage does not affect risk, but liver damage is more likely to occur in continuous rather than being drinkers.
- Gender - The incidence of alcoholic liver disease is increasing in women, who often conceal alcohol misuse and have higher blood ethanol levels than men after drinking because their body mass is lower.
- Genetic - Alcoholism is more common in monozygotic than dizygotic twins. However, polymorphisms in the genes involved in alcohol metabolism, such as aldehyde dehydrogenase (ALD), have yet to be linked to alcoholic liver disease.
- Nutrition - Animals given a choline-deficient diet are more likely to develop alcoholic liver disease.

Pathophysiology

Alcohol is metabolised almost exclusively by the liver via one of two main pathways.

- Eighty percent of alcohol is metabolised to acetaldehyde by the mitochondrial enzyme, alcohol dehydrogenase (ADH). Acetaldehyde forms adducts with cellular proteins in hepatocytes which activate the immune system, leading to cell injury. Acetaldehyde is then metabolised to acetyl-CoA and acetate by ALD. This generates NADH from NAD (nicotinamide adenine dinucleotide), which changes the redox potential of the cell.
- The remaining 20% is metabolised by the mixed function oxidase enzymes of the smooth endoplasmic reticulum. Cytochrome CYP2E1 is an enzyme which oxidise ethanol to acetate. It is induced by alcohol, and during metabolism of ethanol it releases oxygen free radicals, leading to lipid

peroxidation which can induce mitochondrial damage. The CYP2E1 enzyme also metabolises acetaminophen and hence chronic alcoholics are more susceptible to hepatotoxicity from low doses of paracetamol.

It is thought that pro-inflammatory cytokines may also be involved in inducing hepatic damage in alcoholic hepatitis, since endotoxin is released into the blood because of increased gut permetability, leading to release of TNF-α, IL-1, IL-2 and IL-8 from immune cells. All of these cytokines have been implicated in the pathogenesis of liver fibrosis.

In about 80% of patients with severe alcoholic hepatitis, cirrhosis will coexist at presentation. Iron deposition is common and does not necessarily indicate haemochromatosis. The histological features of alcoholic liver disease, which are identical to those of non-alcoholic steatohepatitis.

Clinical features

Alcoholic liver disease has a wide clinical spectrum ranging from mild abnormalities of LFTs on biochemical testing to advanced cirrhosis. The liver is often enlarged in alcoholic liver disease even in the presence of cirrhosis. Peripheral stigmata of chronic liver disease, such as palmar erythema, are more common in alcoholic cirrhosis than in cirrhosis of other aetiologies. Alcohol misuse may also cause damage of other organs and this should be specifically. Three types of alcoholic liver disease are recognised, but in reality these overlap considerably, as do the pathological changes seen in the liver.

Pathological features of alcoholic liver disease—

- Alcoholic hepatitis
- Macrovesicular steatosis
  Lipogranuloma
- Fibrosis and cirrhosis
  Neutrophil infiltration
- Central hyaline sclerosis
  Mallory's hyaline
  Pericellular fibrosis

23.48 Clincial syndromes of alcoholic liver disease—

Fatty liver

- Asymptomatic abnormal
- Normal/large liver
  liver biochemistry

Alcoholic hepatitis

- Jaundice
- Features of portal
- Malnutrition hypertension (e.g. ascites,
- Hepatomegaly
  encephalopathy)

Cirrhosis

- Stigmata of chronic liver
- Large, normal or small liver disease
- Hepatocellular carcinoma
- Ascites/varices/ encephalopathy

Alocoholic fatty liver disease (AFLD) - This usually presents with elevated transaminases in the absence of hepatomegaly. It has a good prognosis and steatosis usually disappears after 3 months of abstinence.

Alocoholic hepatitis - This presents with jaundice and hepatomegaly; complications of portal hypertension may also be present. It has a significantly worse prognosis than AFLD. About one-third of patients die in the acute episode, particularly those with hepatic encephalopathy or a prolonged prothrombin time. Cirrhosis often coexists; if not present, it is the likely outcome if drinking continues. Patients with acute alcoholic hepatitis often deteriorate during the first 1-3 week in hospital. Even if they abstain, it may take up to 6 months for jaundice to resolve. In patients presenting with jaundice who subsequently abstain, the 3-and 5-year survival is 70%. In contrast, those who continue to drink have 3-and 5-year survival rates of 60% and 34% respectively.

Alcoholic cirrhosis—Alcohol-induced cirrhosis often presents with a serious complication such as variceal hemorrhage or ascites, and only half of such patients will survive 5 years from presentation. However, most who survive the initial illness and who become abstinent will survive beyond 5 years.

Ancillary Investigations

Investigations aim to establish alcohol misuse, exclude alternative or coexistent causes of liver disease, such as hepatitis C or haemochromatosis, and assess the severty of liver disease. The clinical history from patient, relatives and friends is most improtant in establishing alcohol misuse, duration and soverity. Biological markers, particularly macrocytosis in the absence of anaemia, may suggest and support a history of alcohol misuse. A raised CGT is not specific for alcohol misuse and will be elevated in the presence of hepatic steatosis and fibrosis. The lvel may not therefore retun to normal with abstinence if chronic liver disease is present. Unexplained ribfractures, particularly bilateral, on a chest X-ray are also suggestive of alcohol misuse. The presence of jaundice suggests alcoholic hepatitis. Determining the extent of liver damage often requires a liver biopsy.

Prothrombin time and bilirubin are used to give a 'discriminant function' (DF), also known as the Maddrey score, which enables the clinician to assess prognosis in alcoholic hepatitis.

Nonalcoholic Fatty Liver Disease (NAFLD)

NAFLD includes simple fatty infiltration (a benign condition called fatty liver) and nonalcoholic steatohepatitis (NASH), a less common but more important variant. NASH (sometimes called steatonecrosis) is diagnosed most often in patients between 40 and 60 years but can occur in all age groups. Many affected patients have obesity, type 2 diabetes mellitus (or glucose intolerance), dyslipidemia, and/or metabolic syndrome.

Pathophysiology of NAFLD

- Fatty liver develops for many reasons, involves many different biochemical mechanisms, and causes different types of liver damage.
- Pathophysiology involves fat accumulation (steatosis), inflammation, and, variably, fibrosis. Steatosis results from hepatic triglyceride accumulation.
- Possible mechanisms for steatosis include reduced synthesis of very low density lipoprotein (VLDL) and increased hepatic triglyceride synthesis (possibly due to decreased oxidation of fatty acids or increased free fatty acids being delivered to the liver).
- Inflammation may result from lipid peroxidative damage to cell membranes. These changes can stimulate hepatic stellate cells, resulting in fibrosis. If advanced, NASH can cause cirrhosis and portal hypertension.

Symptoms and Signs of NAFLD

- Most patients are asymptomatic. However, some have fatigue, malaise, or right upper quadrant abdominal discomfort.
- Hepatomegaly develops in about 75% of patients.
- Splenomegaly may develop if advanced hepatic fibrosis is present and is usually the first indication that portal hypertension has developed.
- Patients with cirrhosis due to NASH can be asymptomatic and may lack the usual signs of chronic liver disease.

Ancillary investigations

- History (presence of risk factors, absence of excessive alcohol intake)
- Serologic tests that rule out hepatitis B and C
- Ultrasound evidence of steatosis or MR elastography with fat fraction

Liver biopsy

- The diagnosis of NASH should be suspected in patients with metabolic syndrome (typically obesity, type 2 diabetes mellitus or a high fasting plasma glucose level, hypertension, and dyslipidemia) and in patients with unexplained laboratory abnormalities suggesting liver disease.
- Differentiating simple steatosis from NASH can be difficult and elevated liver enzymes are not a sensitive predictor for identifying NASH.
- The presence of metabolic syndrome as well as elevated ferritin increases the likelihood that

a patient has NASH rather than simple steatosis.

- When liver enzymes are elevated the most common laboratory abnormalities are elevations in aminotransferase levels.
- Unlike in alcohol-related liver disease, the ratio of aspartate aminotransferase (AST)/alanine aminotransferase (ALT) in NASH is usually < 1.
- Alkaline phosphatase and gammaglutamyl transpeptidase (GGT) occasionally increase.
- Hyperbilirubinemia, prolongation of prothrombin time (PT), and hypoalbuminemia are uncommon.
- For diagnosis, strong evidence (such as a history corroborated by friends and relatives) that alcohol intake is not excessive (eg, is < 20 g/day) is needed,
- Serologic tests should show absence of hepatitis B and C (ie, hepatitis B surface antigen and hepatitis C virus antibody should be negative).
- Liver biopsy reveals damage similar to that seen in alcoholic hepatitis, usually including large fat droplets (macrovesicular fatty infiltration) as well as pericellular or "chicken wire" fibrosis.
- Indications for biopsy include unexplained signs of portal hypertension (eg, splenomegaly, cytopenia) and unexplained elevations in aminotransferase levels that persist for > 6 months in a patient with diabetes, obesity, or dyslipidemia.
- Liver imaging tests, including ultrasonography, CT, and particularly MRI, may identify hepatic steatosis.
- Noninvasive measures of fibrosis such as transient elastography (a test that uses both ultrasound and low-frequency elastic waves), ultrasound elastography, or MR elastography can assess severity of steatosis as well as estimate fibrosis, thereby obviating the need for liver biopsy in many cases
- Transient elastography and ultrasound elastography can be limited by body habitus (too big/fat for ultrasound waves to penetrate adequately), whereas MR elastography is not. However, these tests cannot identify the inflammation typical of NASH and cannot differentiate NASH from other causes of hepatic steatosis.

Hepatitis

Hepatitis is a common cause of jaundice and must be considered in anyone presenting with hepatitic liver blood tests (high transaminases).

All these viruses cause illnesses with similar clinical and pathological features and which are frequently anicteric or even a symptomatic. They differ in their tendency to cause acute and chronic infections.

Causes of viral hepatitis

Common—

- Hepatitis A
- Hepatitis B ± hepatitis D
- Hepatitis C
- Hepatitis E

Less common—

- Cytomegalovirus
- Epstein-Barr virus

Rare—

- Herpes simplex
- Yellow fever

Features of the main hepatitis viruses

	Hepatitis A	Hepatitis B	Hepatitis C	Hepatitis D	Hepatitis E
Virus					
Group	Enterovirus	Hepadna virus	Flavivirus	Incomplete virus	Calicivirus
Nucleic acid	RNA	DNA	RNA	RNA	RNA
Size (diameter)	27 non	42 nm	30-38 nm	35 nm	27 nm

Incubation (weeks)	2-4	4-20	2-26	6-9	3-8
Spread faeces	Yes	No	No	No	Yes
Blood	Uncommon	Yes	Yes	Yes	No
Saliva	Yes	Yes	Yes	Unknown	Unknown
Sexual	Uncommon	Yes	Uncommon	Yes	Unknown
Vertical	No	Yes	Uncommon	Yes	No
Chronic infection	No	Yes	Yes	Yes	No
Prevention					
Active	Vaccine	Vaccine	No	Prevented by hepatitis	No
Passive	Immune serum globulin	Hyperimmune serum globulin	No	B vaccination	No

Note : All body fluids are potentially infectious, although some (i.g. urine) are less infectious than others.

Clinical features of acute infection

Non-specific prodromal illness characterised by headache, myalgia, arthralgia, nausea and anorexia usually precedes the development of jaundice by a few days to 2 weeks. Vomiting and diarrhoea may follow, and abdominal discomfort is common. Dark urine and pale stools may precede jaundice. There are usually few physical signs. The liver is often tender but only minimally enlarged. Occasionally, mild splenomegaly and cervical lymphadenopathy are seen. These are more frequent in children or those with Epstein-Barr virus infection.

Jaundice may be mild and the diagnosis may be suspected only after finding abnormal liver blood tests. Symptoms rarely last longer than 3-6 weeks.

Complications of acute viral hepatitis

- Acute liver failure
- Chronic liver disease and cirrhosis (hepatitis B and C)
- Cholestatic hepatitis (hepatitis A)
- Relapsing hepatitis
- Aplastic anaemia

Investigations

A hepatitic pattern of Liver function tests develops, With serum transaminases typically between 200 and 2000 U/L. The plasma bilirubin reflects the degree of liver damage. The ALP rarely exceeds twice the upper limit of normal. Prolongation of the prothrombin time indicates the severity of the hepatitis but, except in rare cases of acute liver failure, rarely exceeds 25 seconds. The white cell count is usually normal with a relative lymphocytosis. Serological tests confirm the aetiology of the infection.

Hepatitis A

The hepatitis A virus (HAV) belongs to the picornavirus group of enteroviruses. HAV is highly infectious and is spread by the faecal-oral route. Infected individuals, who may be a symptomatic, excrete the virus in faeces for about 2-3 weeks before the onset of symptoms and then for a further 2 weeks or so. Infection is common in children but often a symptomatic, and so up to 30% of adults will have serological evidence of post infection but give no history of jaundice. Infection is also more common in areas of overcrowding and poor sanitation. In occasional outbreaks water and shellfish have been the vehicles of transmission.

In contrast to hepatitis B, a chronic carrier state does not occur.

Investigations

Only one HAV antigen has been found; individuals infected with HAV make an antibody to this antigen (anti-HAV). Anti-HAV is important in diagnosis, as HAV is only present in the blood transiently during the incubation period. Excretion in the stools occurs for only 7-14 days after the onset of the clinical illness and the virus cannot be grown readily. Anti-HAV of the IgM type indicating a primary immune response, is already present in the blood at the onset of the clinical illness and is diagnostic of an acute HAV infection. Titres of this anti-body fall to low levels within about 3 months of recovery. Anti-HAV infection is common and this antibody persists for year after infection, but it can be used as a marker of previous HAV infection. Its presence indicates immunity to HAV.

Hepatitis B

The hepatitis B virus consists of a core containing DNA and a DNA polymerase enzyme needed for virus replication. The core of the virus is sorrounded by surface protein. The virus, also called a Dane particle, and an excess of its surface protein (known as hepatitis B surface antigen) circulate in the blood. Humans are the only source of infection.

Hepatitis B infection affects 300 million people and is one of the most common cuase of chronic liver disease and hepatocellular carcinoma world-wide.

Hepatitis B may cause an acute viral hepatitis; however, the acute infection is often asymptomatic particularly when acquired at birth. Many individuals with chronic hepatitis B are also asymptomatic. Chronic hepatitis, associated with elevated serum transaminases, may occur and can lead to cirrhosis, usually after decades of infection.

The risk of progression to chronic liver disease depends on the source of infection. Vertical transmission from mother to child in the perinatal period is the most common cause of infection world-wide and carries the highest risk.

In this setting, adaptive immune responses to HBV may be absent initially, with apparent immunological tolerance. Several mechanisms contribute towards this tolerance.

- Firstly, the introduction of antigen in the neonatal period is tolerogenic.
- Secondary, the presentation of such antigen within the liver, as described above, promotes tolerance; this is particularly evident in the absence of a significant innate or inflammatory response.
- Finally, very high loads of antigen may lead to so called exhaustion of cellular immune responses.

However, the state of tolerance is not permanent and may be reversed as a result of therapy, or through spontaneous changes in innate responses such as interferon-alpha and NK cells, accomparied by host mediated immunopathology.

Investigation

Serology—HBV contains several antigens to which infected persons can make immune responses; these antigens and their antibodies are important in identifying HBV infection .

- *The hepatitis B surface antigen* (HBsAg) is an indicator of active infection, and a negative test for HBsAg makes HBV infection very unlikely. In acute liver failure from hepatitis B and liver damage is mediated by viral clearance and so HBsAg is negative, with evidence of recent infection shown by the presence of hepatitis B core IgM. HBsAg appears in the blood late in the incubation period but before the prodromal phase of acute type B hepatitis, it may be present for a few days only disappearing even before jaundice has developed, but usually last for 3-4 weeks and can persist for up to 5 months indicates chronic infection. Antibody to HBsAg (anti-HBs) usually appears after about 3-6 months and persists for many years or perhaps permanently. Anti-HBs implies either a previous infection, in which case anti-HBc is usually also present, or previous vaccination, in which case anti HBc is not present.

- *The hepatitis B core antigen* (HBcAg) is not found in the blood, but antibody to it (anti-HBc) appears early in the illness and rapidly reaches a high titre, which subsides gradually but then persists. Anti-HBc is initially of IgM type with IgG antibody appearing later. Anti-HBc (IgM) can sometimes reveal an acute HBV infection when the HBsAg has disappeared and before anti-HBs has developed.
- *The hepatitis B e antigen* (HBeAg) is an indicator of viral replication. In acute hepatitis B it may appear only transiently at the outset of the illness; its apearance is followed by the production of antibody (anti-HBe). The HBeAg reflects active replication of the virus in the liver.

Chronic HBV infection is marked by the presence of HBsAg or anti-HBc (IgG) in the blood. Usually, HBeAg or anti-HBe is also present; HBeAg indicates continued active replication of the virus in the liver. Although the presence of anti-HBe usually implies low viral replication, the exception is HBeAb-positive replicating chronic hepatitis B (also called 'pre-core mutant infection), in which high levels of serum HBV-DNA are seen, despite negative HBeAg.

Viral load—HBV-DNA can be measured by polymerase chain reaction (PCR) in the blood. Viral loads are usually in excess of 10^5 copies/mL in the presence of active viral replication, as indicated by the presence of e antigen. In contrast, in those with low viral replication, HBsAg and anti-HBe-positive, viral loads are less than 10^5 copies mL. The exception is in patients who have a mutation in the pre-core protein, which means they cannot secrete antigen into serum. Such individuals will be anti-HBe-positive but have a high viral load and often evidence of chronic hepatitis. These mutations are common in the Far East and those patients affected are classified as having e antigen-negative chronic hepatitis. They respond differently to antiviral drugs from those with classical e antigen-positive chronic hepatitis.

Measurement of viral load is important in monitoring antiviral therapy and identifying patients with pre-core mutants. Specific HBV genotypes can also be identified using PCR. Genotypes B and C appear to have more aggressive disease that responds less well to interferon.

Hepatitis D (Delta virus)

The hepatitis D virus (HDV) is an RNA-defective virus which has no independent existence; it requires HBV for replication and has the same sources and modes of spread. It can infect individuals simultaneously with HBV, or can superinfect those who are already chronic carriers of HBV Simultaneous infections give rise to acute hepatitis, which is often severe but is limited by recovery from the HBV infection. Infections in individuals who are chronic carriers of HBV can cause acute hepatitis with spontaneous recovery, and occasionally simultaneous cessation of the chronic HBV infection occurs. Chronic infection with HBV and HDV can also occur, and this frequently causes rapidly progressive chronic hepatitis and eventually cirrhosis.

HDV has a world-wide distribution. It is endemic in arts of the Mediterranean basin, Africa and South America, where transmission is mainly by close personal contact and occasionally by vertical transmission from mothers who also carry HBV. In no-endemic areas, transmission is mainly a consequence of parenteral drug misuse.

Investigations

HDV contains a single antigen to which infected individuals make an antibody (anti-HDV). Delta antigen appears in the blood only transiently, and in practice diagnosis depends on detecting anti-HDV. Simultaneous infection with HBV and HDV followed by full recovery is associated with the appearance of low titers of anti-HDV of IgM type within a few days of the onset of the illness. This antibody generally disappears within 2 months but persists in a few patients. Superinfection of patients with chronic HBV infection leads to the production of high titers of anti-HDV, initially IgM and later IgG. Such patients may then develop chronic infection with both viruses, in which case anti-HDV titres plateau at high levels.

Hepatitis C

This is caused RNA flavivirus. Acute symptomatic infection with hepatitis C is rare. Most individuals are unaware of when they became infected and only identified when they develop chronic liver disease.

Eighty per cent of individuals exposed to the virus become chronically infected and late spontaneous viral clearance is rare.

Hepatitis C is the cause of what used to be known as 'non-A, non-B hepatitis', a syndrome of acute hepatitis often with jaundice seen after a transfusion of blood or blood products. Following the identification of the virus In 199Q, blood donors are now screened for infection in many parts of the world. New cases of post- transfusion hepatitis C no longer occur in the UK.

Hepatitis C infection is usually now identified in asymptomatic individuals screened because they have risk factors for infection such as previous injection drug use or they have incidentally been found to have abnormal liver blood tests. Although most individuals remain asymptomatic until progression to cirrhosis occurs, fatigue can complicate chronic infection and appears to be unrelated to the degree of liver damage.

Investigations

Serology and virology—The HCV protein contains several antigens that give rise to antibodies in an infected person, and these are used in diagnosis. It may take 6-12 weeks for antibodies to appear in the blood following acute infection such as a needlestick injury. In these cases hepatitis C RNA can be identified in the blood as early as 2-4 weeks after infection. Active infection is confirmed by the presence of serum hepatitis C RNA in anyone who is antibody-positive. Anti-HCV antibodies persist in serum even after viral clearance, whether spontaneous or post-treatment.

Molecular analysis—There are six common viral genotypes whose distribution varies worldwide. Genotype has no effect on progression of liver disease but does affect response to treatment. Genoype 1 is most common in northern Europe and is less easy to eradicate with current treatment.

Risk factors for the acqusition of chronic hepatitis C infection—

- Intravenous drug misuse (95% of new cases in the UK)
- Unscreened blood products
- Vertical transmission (3% risk)
- Needlestick injury (3% risk)
- Iatrogenic parenteral transmission, (i.e. contaminated vaccination needles)
- Sharing toothbrushes/razors

Liver function tests—LFTs may be normal or show fluctuating serum transaminases between 50 and 200 U/L. Jaundice is rare and only usually appears in end-stage cirrhosis.

Liver histology—Serum transaminase levels in hepatitis C are a poor predictor of the degree of liver fibrosis, and so a liver biopsy is often required to stage the degree of liver damage. Non-invasive methods of assessing liver fibrosis in hepatitis C infection remain an area of active research. The degree of inflammation and fibrosis can be scored histologically. The most common scoring system use din hepatitis C is the Metavir system, which scores fibrosis from 1 to 4, the latter equating to cirrhosis.

Hepatitis E

Hepatitis E is caused by an RNA virus which is endemic in India and the Middle East. An increase in prevalence has recently been noted in northern Europe and infection is no longer seen only in travellers from an endemic area.

The clinical presentation and management of hepatitis E are similar to that of hepatitis A. Disease is spread via the faecal-oral route; in most cases, it presents as a self-limiting acute hepatitis and does not cause chronic liver disease.

Hepatitis E differs from hepatitis A in that infection during pregnancy is associated with the development of acute liver failure, which has a high mortality. In acute infection, IgM antibodies to HEV are positive.

Other forms of viral hepatitis

Non-A, non-B, non-C (NANBNC) or non-A-E hepatitis is the term used to describe hepatitis thought to be due to a virus that is not HAV, HBV, HCV or HEV. Other viruses which affect the liver probably do exist, but the hepatitis viruses described above now account for the majority of hepatitis virus infections. Cytomegalovirus and Epstein-Barr virus infection causes abnormal LFTs in most patients, and occasionally icteric hepatitis occurs. Herpes simplex is a rare cause of hepatitis in adults, and most of these patients are immuno-compromised. Abnormal LFTs are also common in chickenpox, measles, rubella and acute HIV infection.

Causes of abnormal liver blood tests in HIV infection—

Hepatitic blood tests—

- Chronic hepatitis C
- Antiretroviral drugs
- Chronic hepatitis B
- Cytomegalovirus

Cholestatic blood tests—

- Tuberculosis
- Sclerosing cholangitis due to Cryptosporidia
- Atypical mycobacterium

Jaundice

1. Jaundice is usually detectable clinically when the plasma bilirubin exceeds 50 mol/L (-3 mg/dL).
2. The causes of jaundice overlap with the causes of abnormal LFTs.
3. In a patient with jaundice it is useful to consider whether the cause might be pre-hepatic, hepatic or post-hepatic.

Common causes of elevated serum transaminases

Minor elevation (< 100 U/L)

- Chronic hepatitis C
- Haemochromatosis
- Chronic hepatitis B
- Fatty liver disease

Moderate elevation (100-300 U/L)

As above plus:

- Alcoholic hepatitis
- Autoimmune hepatitis
- Non-alcoholic steatohepatitis
- Wilson's disease

Major elevation (> 300 U/L)

- Drugs (e.g. paracetamol)
- Ischaemic liver
- Acute viral hepatitis
- Toxins (e.g. *Amanita phalloides* poisoning)
- Autoimmune liver disease
- Flare of chronic hepatitis B

Pre-Hepatic Jaundice

1. This is caused either by haemolysis or by congenital hyperbilirubinaemia, and is characterised by an isolated raised bilirubin level.
2. In haemolysis, destruction of red blood cells or their precursors in the marrow causes increased bilirubin production.
3. Jaundice due to haemolysis is usually mild because a healthy liver can excrete a bilirubin load six times greater than normal before unconjugated bilirubin accumulates in the plasma.
4. The newborn, who have a reduced capacity to metabolise bilirubin .Hence this rule does not apply.
5. The most common form of non-haemolytic hyperbilirubinaemia is Gilbert's syndrome, an inherited disorder of bilirubin metabolism
6. Other inherited disorders of bilirubin metabolism also exist but they are very rare.

Congenital non-haemolytic hyperbilirubinaemia

Syndrome	Inheritance	Abnormality	Clinical features/treatment
Unconjugated hyperbilirubinaemia			
Gilbert's	Autosomal dominant	Glucuronyl transferase Bilirubin uptake	Mild jaundice, especially with fasting No treatment necessary
Crigler-Najjar			
Type I	Autosomal recessive	Absent glucuronyl transferase	Rapid death in neonate (kernicterus)
Type II	Autosomal dominant	Absent glucuronyl Glucuronyl transferase	Presents in neonate Phenobarbital, ultraviolet light or liver transplant as treatment
Conjugated hyperbilirubinaemia			
Dubin-Johnson	Autosomal recessive	Canalicular excretion of organic anions, including bilirubin	Mild No treatment necessary
Rotor's	Autosomal recessive	Bilirubin uptake Intrahepatic binding	Mild No treatment necessary

Hepatocellular Jaundice

1. Hepatocellular jaundice results from an inability of the liver to transport bilirubin across the hepatocyte into the bile, occurring as a consequence of parenchymal liver disease.
2. Bilirubin transport across the hepatocytes may be impaired at any point between uptake of unconjugated bilirubin into the cells and transport of conjugated bilirubin into the canaliculi.
3. The swelling of cells and oedema resulting from the disease itself may cause obstruction of the biliary canaliculi.
4. In hepatocellular jaundice the concentrations of both unconjugated and conjugated bilirubin in the blood increase.
5. Hepatocellular jaundice can be due to acute or chronic liver injury.
6. AST- ALT; GGT, ALP : Characteristically, jaundice due to parenchyma liver disease is associated with increases in transaminases (AST, ALT), but increases in other LFTs, including cholestatic enzymes (GGT, ALP) may occur, and suggest specific aetiologies.
7. AST > 1000 U/L : Acute jaundice in the presence of AST > 1000 U/L is highly suggestive of an infectious cause (e.g. hepatitis A, B), drugs (e.g. paracetamol) or hepatic ischaemia.
8. Imaging : Imaging is essential, in particular to identify features suggestive of cirrhosis (e.g. irregular liver outline, splenomegaly), to define the patency of hepatic arteries and veins, and of the portal vein, and to obtain evidence of portal hypertension.
9. Liver biopsy : Liver biopsy has an important role in defining the aetiology of hepatocellular jaundice and the extent of established liver injury.

Obstructive (Cholestatic) Jaundice

Cholestatic jaundice may be caused by:

- Failure of hepatocytes to initiate bile flow
- Obstruction of bile flow in the bile ducts or portal tracts
- Obstruction of bile flow in the extrahepatic bile ducts between the porta hepatis and the papilla of Vater.
- In the absence of treatment, cholestatic jaundice tends to become progressively more

severe because conjugated bilirubin is unable to enter the bile canaliculi and passes back into the blood, and also because there is a failure of clearance of unconjugated bilirubin arriving at the liver cells.

Causes *of* cholestatic jaundice

Intrahepatic

- Primary biliary cirrhosis
- Pregnancy
- Primary sclerosing cholangitis
- Inherited cholestatic liver disease, e.g. benign recurrent intrahepatic cholestasis
- Alcohol
- Drugs
- Cystic fibrosis
- Chronic right heart failure
- Severe bacterial infections
- Hepatic infiltrations (lymphoma, granuloma, amyloid, metastases)

Extrahepatic

- Carcinoma
  Ampullary
  Pancreatic
  Bile duct
  (cholangiocarcinoma)
- Choledocholithiasis
- Parasitic infection
- Traumatic biliary strictures
- Chronic pancreatitis

Liver metastases

- Cholestasis may result from defects at more than one of these levels.
- Those confined to the extrahepatic bile ducts may be amenable to surgical correction.

Clinical features in cholestatic jaundice

Obstruction of the bile duct drainage due to blockage of the extrahepatic biliary tree :

- Pale stools
- Dark urine
- Pruritus accompanied by skin excoriations.
- Peripheral stigmata of chronic liver disease are absent.

Courvoisier's Law

- If the gallbladder is palpable, the jaundice is unlikely to be caused by biliary obstruction due to gallstones, probably because a chronically inflamed stone-containing gallbladder cannot readily dilate. This is Courvoisier's Law, and suggests that jaundice is due to a malignant biliary obstruction (e.g. pancreatic cancer).

Charcot's triad

- Cholangitis is characterised by 'Charcot's triad' of jaundice, right upper quadrant pain and fever.

ALP and GGT

- Cholestatic jaundice is characterised by a relatively greater elevation of ALP and GGT than the aminotransferases.

Ultrasound

- Ultrasound evaluation is indicated in all cases to determine whether there is evidence of mechanical obstruction and dilatation of the biliary tree .

Clinical features and complications of cholestatic jaundice

Cholestasis
- Jaundice
- Dark urine

Early features
- Pale stools
- Pruritus

Late features :
- Malabsorption (vitamins A, D, E and K)
  Weight loss
  Steatorrhoea
  Osteomalacia
  Bleeding tendency

Cholangitis
- Fever
- Pain (if gallstones are present)
- Rigors

Clinical features suggesting an underlying cause of cholestatic jaundice

Clinical feature	Causes
Jaundice Static or increasing	Carcinoma Primary biliary cirrhosis Primary sclerosing cholangitis
Fluctuating	Choledocholithiasis Stricture

	Pancreatitis Choledochal cyst Primary sclerosing cholangitis
Abdominal pain	Choledocholithiasis Pancreatitis Choledochal cyst
Cholangitis	Stone Stricture Choledochal cyst
Abdominal scar	Stone Stricture
Irregular hepato-megaly	Hepatic carcinoma
Palpable gallbladder	Carcinoma below cystic duct (usually pancreas)
Abdominal mass	Carcinoma Pancreatitis (cyst) Choledochal cyst
Occult blood in stools	Ampullary tumour

Ascites

1. Excess accumulation of free fluid in the peritoneal cavity is known as Ascites

Causes of ascites

Common causes

Malignant disease
Hepatic
Peritoneal
- Cardiac failure
- Hepatic cirrhosis

Other causes

Hypoproteinaemia
- Pancreatitis

Nephrotic syndrome
- Lymphatic obstruction

Protein-losing enteropathy
Malnutrition
- Infection Tuberculosis

Hepatic venous occlusion
Budd-Chiari syndrome
Veno-occlusive disease

Rare causes

- Meigs syndrome
- Hypothyroidism

('Meigs' syndrome is the association of a right pleural effusion with or without ascites and a benign ovarian tumour. The ascites resolves on removal of the tumour)

Pathophysiology

Splanchnic vasodilatation is thought to be the main factor leading to ascites in cirrhosis.

↓

This is mediated by vasodilators (mainly nitric oxide) that are released when portal hypertension causes shunting of blood into the systemic circulation.

↓

Systemic arterial pressure falls due to pronounced splanchnic vasodilatation as cirrhosis advances.

↓

This leads to activation of the renin-angiotensin system with secondary aldosteronism, increased sympathetic nervous activity, increased atrial natriuretic hormone secretion and altered activity of the kallikrein-kinin system

↓

These systems tend to normalise arterial pressure but produce salt and water retention.

↓

The combination of splanchnic arterial vasodilatation and portal hypertension alters intestinal capillary permeability,

↓

Promoting accumulation of fluid within the peritoneum

↓

Ascites

Clinical features

1. Small amounts of ascites are asymptomatic
2. Larger accumulations of fluid (> 1 L) produces clinical manifestations
   - Abdominal distension
   - fullness in the flanks

   Shifting dullness
   - Shifting dullness on percussion

   Fluid thrill
   - When the ascites is marked, a fluid thrill is present
3. In obese patients, much larger volumes of

ascites may accumulate before they are detectable clinically.

Other physical features

- Distortion or eversion of the umbilicus
- Herniae
- Abdominal striae
- Divarication of the recti
- Scrotal oedema.
- Dilated superficial abdominal veins may be seen if the ascites is due to portal hypertension.

Investigations

Ultrasonography

1. Ultrasonography is the best means of detecting ascites, particularly in the obese and those with small volumes of fluid

Paracentesis

2. Paracentesis (if necessary under ultrasonic guidance) can also be used to confirm the presence of ascites but is most useful for obtaining ascitic fluid for analysis.

Ascites fluid appearance and analysing

Cause/appearance

- Cirrhosis: clear, straw-coloured or light green
- Malignant disease: bloody
- Infection: cloudy
- Biliary communication: heavy bile staining
- Lymphatic obstruction: milky-white (chylous)

Useful investigations

- Total albumin (plus serum albumin) and protein
- Amylase
- Neutrophil count
- Cytology
- Microscopy and culture

Radiography

- Pleural effusions are found in about 10% of patients, usually on the right side (hepatic hydrothorax); most are small and only identified on chest X-ray, but occasionally a massive hydrothorax occurs.
- Pleural effusions, particularly those on the left side, should not be assumed to be due to the ascites.

Serum-Acites Albumin Gradient (SAAG)

- Measurement of the protein concentration and the serum-ascites albumin gradient (SAAG) are used to distinguish a transudate from an exudate.

Transudate Ascites

- Cirrhotic patients typically develop a transudate with a total protein concentration below 25 g/L and relatively few cells.
- However, in up to 30%, of patients, the total protein concentration is more than 30 g/L.
- In these cases it is useful to calculate the SAAG by subtracting the concentration of the ascites fluid albumin from the serum albumin.
- A gradient of more than 11 g/L is 96% predictive that ascites is due to portal hypertension.
- Venous outflow obstruction due to cardiac failure or hepatic venous outflow obstruction can also cause a transudative ascites, as indicated by an albumin gradient above 11 g/L, but unlike in cirrhosis the total protein content is usually- above 25g/L.

Exudate Ascites

- Ascites protein concentration above 25 g/ L or a SAAG of less than 11 g/ L) raises the possibility of infection (especially tuberculosis), malignancy, hepatic venous obstruction, pancreatic ascites or, rarely, hypothyroidism.

Pancreatic ascites

- Ascites amylase activity above 1000 U/L identifies pancreatic ascites

Ascites glucose concentrations

- Low ascites glucose concentrations suggest malignant disease or tuberculosis.

Cytological examination

- Cytological examination may reveal malignant cells (onethird of cirrhotic patients with a bloody tap have a hepatoma).

Full blood count

- Polymorphonuclear leucocyte counts above $250 \times 10^6/L$ strongly suggest infection (spontaneous bacterial peritonitis).

Laparoscopy

- Laparoscopy can be valuable in detecting peritoneal disease.

Abdominal pain

1. Abdominal pain results from GI disease and extraintestinal conditions involving the genitourinary tract, abdominal wall, thorax, or spine.
2. The most common causes of abdominal pain are irritable bowel syndrome and functional dyspepsia

There are four types of abdominal pain

Visceral pain

- Visceral pain generally is midline in location and vague in character.
- Gut organs are insensitive to stimuli such as burning and cutting but are sensitive to distension, contraction, twisting and stretching.
- Pain from unpaired structures is usually but not always felt in the midline.
- Noninflammatory visceral sources include mesenteric ischemia and neoplasia.

Parietal pain

- It is localized and precisely described .i.e. sharp and lateralised pain
- The parietal peritoneum is innervated by somatic nerves, and its involvement by inflammation, infection or neoplasia
- Common inflammatory diseases with pain include peptic ulcer, appendicitis, diverticulitis, inflammatory bowel disease, and infectious enterocolitis.

Referred pain

- Gallbladder pain -is referred to the back or shoulder tip
- Pancreatitis- The pain radiates directly through the abdomen to the back in approximately one half of cases.

Psychogenic

- Cultural, emotional and psychosocial factors influence everyone's experience of pain.
- In some patients, no organic cause can be found despite investigation, and psychogenic causes (depression or somatisation disorder) may be responsible.

The acute abdomen

This accounts for approximately 50% of all urgent admissions to general surgical units. The acute abdomen is a consequence of one or more pathological processes.

Inflammation

- Pain develops gradually, usually over several hours.
- It is initially rather diffuse until the parietal peritoneum is involved, when it becomes localised.
- Movement exacerbates the pain
- Abdominal rigidity and guarding occur

Perforation

- When a viscus perforates, pain starts abruptly
- It is severe and leads to generalised peritonitis (guarding and rebound tenderness with rigidity)

Obstruction

- Pain is colicky, with spasms which cause the patient to writhe around and double up.
- Colicky pain which does not disappear between spasms suggests complicating inflammation.

Investigations

Blood :

- Full blood count and Urea and electrolytes and amylase taken to look for evidence of dehydration, leucocytosis and pancreatitis.

Erect chest X-ray

- An erect chest X-ray may show air under the diaphragm suggestive of perforation
- Plain abdominal film may show evidence of obstruction

Abdominal ultrasound

- An abdominal ultrasound may help if gall stones or renal stones are suspected.
- It is also useful in the detection of free fluid and any possible intra-abdominal abscess.

Contrast studies

- Contrast studies, by either mouth or anus, are useful in the further evaluation of intestinal obstruction
- Essential in the differentiation of pseudo-obstruction from mechanical large bowel obstruction.

CT abdomen

- To seek an evidence of pancreatitis, retroperitoneal collections or masses, including an aortic aneurysm)

Angiography

To detect mesenteric ischaemia

Diagnostic laparotomy

It should be considered when the diagnosis has not been revealed by other investigations.

Note- All patients must be carefully and regularly reassessed (every 2-4 hours) so that any change in condition which might alter both the suspected diagnosis and clinical decision can be observed and acted upon early.

Causes of acute abdominal pain

Inflammation

- Appendicitis
- Pancreatitis
- Diverticulitis
- Pyelonephritis
- Cholecystitis
- Intra-abdominal abscess
- Pelvic inflammatory disease

Perforation/rupture

- Peptic ulcer
- Ovarian cyst
- Diverticular disease
- Aortic aneurysm

Obstruction

- Intestinal obstruction
- Ureteric colic
- Biliary colic

MCQs

1. Patients with hepatitis typically have an AST: ALT ratio of at least 2:1. The AST rarely exceeds 300 U/L indicates which hepatitis
   a. Alcoholic hepatitis
   b. Viral hepatitis
   c. Drug toxicity hepatitis
   d. Autoimmune hepatitis
2. Which one is correct?

a. Liver biopsy remains the criterion standard in the evaluation of patients withliver disease, particularly in patients with chronic liver diseases & Liver biopsy is necessary for diagnosis but is more often useful in assessingthe severity (grade) and stage of liver damage, in predicting prognosis, and inmonitoring response to treatment.

b. Liver biopsy is not criterion standard in the evaluation of patients with liver disease, particularly in patients with chronic liver diseases & Liver biopsy is necessary for diagnosis but is more often useful in assessing the severity (grade) and stage of liver damage, in predicting prognosis, and in monitoring response to treatment.

c. Liver biopsy is not necessary for diagnosis but is more often useful in assessing the severity (grade) and stage of liver damage, in predicting prognosis, and in monitoring response to treatment.

d. Liver biopsy is not useful in assessing the severity (grade) and stage of liver damage, in predicting prognosis, and in monitoring response to treatment.

3. Itching occurs with acute liver disease observed in

a. Appearing early in obstructive jaundice (from biliary obstruction or drug-Induced cholestasis), somewhat later in hepatocellular disease (acute hepatitis). Itching can occur inany liver disease, particularly once cirrhosis is present & Itching also occurs in chronic liver diseases, typically the cholestatic formssuch as primary biliary cirrhosis and sclerosing cholangitis.

b. Appearing early in hepatocellular jaundice,
c. Appearing early in haemolytic jaundice
d. Appearing early in haemolytic & hepatocellular jaundice

4. An enlarged left supraclavicular node (Virchow's node) or periumbilical nodule (SisterMary Joseph's nodule) suggests
a. Cholecystitis
b. Viral hepatitis
c. Alcoholic hepatitis
d. Abdominal malignancy

5. Ascending cholangitis and biliary sepsisdiagnosed using which triad
a. Anderson Triad b. Beck's triad
c. Charcot's triad d. Dieulafoy's triad

6. Grey turner sign is seen in
a. Acute cholecystitis b. Acute appendicitis
c. Acute pancreatitis d. Acute hepatitis

7. Which of the following physical findings may indicate severe necrotizing pancreatitis?
a. Jaundice
b. Erythematous skin nodules
c. Fever
d. Hypotension

8. Puddle sign is used to detect
a. Ascites
b. Acute cholecystitis
c. Acute appendicitis
d. Abdominal malignancy

9. Hepatorenal syndrome is diagnosed
a. Creatinine clearance rate of less than 60mL/min is present, Serum creatinine level of greater than 2.5 mg/dL & Urine sodiumlevel of less than 10 mEq/L
b. Creatinine clearance rate of less than 70mL/min is present, Serum creatinine level of greater than 3.5 mg/dL & Urine sodiumlevel of less than 10 mEq/L
c. Creatinine clearance rate of less than 80mL/min is present, Serum creatinine level of greater than 4.5 mg/dL & Urine sodiumlevel of less than 10 mEq/L
d. Creatinine clearance rate of less than 40mL/min is present, Serum creatinine level of greater than 1.5 mg/dL & Urine sodiumlevel of less than 10 mEq/L

Answer

1.	a	2.	a	3.	a
4.	c	5.	a	6.	c
7.	b	8.	a	9.	d